Solution Design: Citizen-Led Data Generation & Management Platform — Wetland Monitoring (Phase 1)

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Scope of this document

This document describes a citizen-led data generation and management platform built under the NBD assignment "Technical Support to Implement Citizen-Led Data Generation and Management Activities". Phase 1 deploys the platform with wetland monitoring as its first thematic use case, at two transboundary pilot sites (Mara and Sio-Siteko).

The platform is designed so that adding a new thematic domain in later phases is a development task, not a re-architecture. The reusable parts are the ingest channels (USSD, WhatsApp, KoboCollect), the admin moderation workflow, the portal shell, the identity and access model, and the external-data ingestion pattern. The domain-specific parts (the form, the scoring rule, the per-domain external sources) are added by a developer through a new form definition, new scoring code, and configuration

1. Background & Context

This document describes the solution design for the data platform built under the assignment “Technical Support to Implement Citizen-Led Data Generation and Management Activities”, commissioned by the Nile Basin Discourse (NBD).

NBD brings together civil society organisations, academic institutions, cultural/religious institutions and government bodies across the Nile Basin to promote equitable, sustainable management of shared water resources. Citizens living around basin ecosystems observe ecological change daily, but that knowledge has no formal channel into management systems.

Phase 1 — wetland monitoring at two pilot sites:

- Mara Basin — Kenya and Tanzania
- Sio-Siteko Basin — Kenya and Uganda

Wetlands are the first domain . Adding a new domain means adding a form, a scoring rule set , optional new external datasets and role assignments . The system will not need architecture level changes

This document describes the technical backbone of that system. The platform collects data submitted by community members from the selected sites, processes and validates it, and presents results to partners and decision-makers through a public-facing portal.

This document covers what the platform can control and measure. Programme-level outcomes — whether authorities act on alerts, whether management plans are adopted, whether partner organisations change behaviour — depend on stakeholder engagement, training, and governance processes that sit outside the platform. Those indicators are tracked by NBD through the programme results matrix and are not in scope here.

1.1 Platform success indicators

The platform is considered successful if it reliably delivers:

1. Pollution reports ingested and published to the portal per basin per month.
2. Sampling submissions received per active site per month.
3. Wetland health classes (A–E) published per site per month.
4. Portal uptime and public API availability within the targets defined below.
5. Portal data accessible to the public without a login.

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2. Problem Statement

Wetland managers and government authorities in the Nile Basin have limited access to reliable, regular data on the health of transboundary wetlands. Formal monitoring is infrequent, expensive, and conducted by specialists who are rarely present on the ground.

The Mara Wetlands form a large papyrus-dominated floodplain of roughly 400–500 km² whose extent varies seasonally. The wetland spans Butiama, Rorya, Tarime, and Serengeti districts and supports an estimated 56,000 people who rely primarily on rain-fed agriculture and small-scale fishing.

It has lost more than 100 km² to land conversion over the past decade. The drivers are upstream deforestation in the Mau Highlands, sedimentation, nutrient loading from intensive cultivation, and pollution from small-scale mining. Climate projections show ≈ 1.8 °C of warming and 10–12% more rainfall by 2050, raising both flood and drought risk for ~56,000 wetland-dependent residents

The Sio-Siteko Wetland Landscape — ~415 km² straddling Kenya and Uganda — is an Important Bird Area with over 520 bird species. It is degrading from unsustainable land use, charcoal-driven vegetation clearing, encroachment, sand harvesting, invasive species (Mimosa pudica, Lantana camara, Pontederia crassipes), and weak transboundary coordination

Reconnaissance field visits under this assignment confirmed those pressures. They also documented that most observed pollution incidents are reported only verbally to local leaders, with no formal record reaching the relevant authority inside an actionable window

Communities alongside the Mara and Sio-Siteko wetlands observe changes daily: unusual water colour, reduced fish catch, encroachment by farms, shifting flood patterns, invasive species, animal loss among others. This knowledge has no formal channel into decision-making systems hence this platform creates that channel.

Current state vs. proposed state

Dimension	Current state	Proposed state
Pollution reporting cadence	Ad hoc; verbal to local leaders; rarely reaches authority in an actionable window	Same-day digital report via USSD or WhatsApp; visible on public portal
Wetland health assessment	Specialist field surveys, annual or rarer; results held in PDF	Monthly citizen sampling at fixed sites; quarterly academic shadow-validation; results published as live scores
Data accessibility for decisions	Held by individual agencies, format varies, often unavailable	Public portal with health scores, trends, pollution-incident map, and downloadable PDF reports
Indigenous knowledge	Captured informally at Monthly Barazas; not aggregated	Captured through a structured FGD form; aggregated into the IK signal that adjusts the composite health score

The design must respect three real-world constraints:

Connectivity. Many sampling sites have no mobile data coverage. The USSD channel requires only GSM voice coverage. KoboCollect stores data offline and syncs when connectivity is available.

Device access. Not all community members have smartphones. The pollution reporting workflow must be accessible from a basic feature phone.

Digital literacy. Citizen reporters and citizen scientists are not technical users. Every interaction must be guided, menu-driven, and completable without reading instructions.

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3. Solution Overview

Akvo will build a mobile-first, citizen-centric data platform. It enables community-driven data collection, administration, and visualisation through familiar and accessible tools.

The platform is a Minimum Viable Product (MVP) — a foundational system supporting the two pilot basins. It can be extended if NBD decides to add new basins, wetlands, or data layers during scale-up.

The platform has three layers: data collection, administration, and visualisation. Two purpose-built workflows feed into a single public-facing wetland data portal.

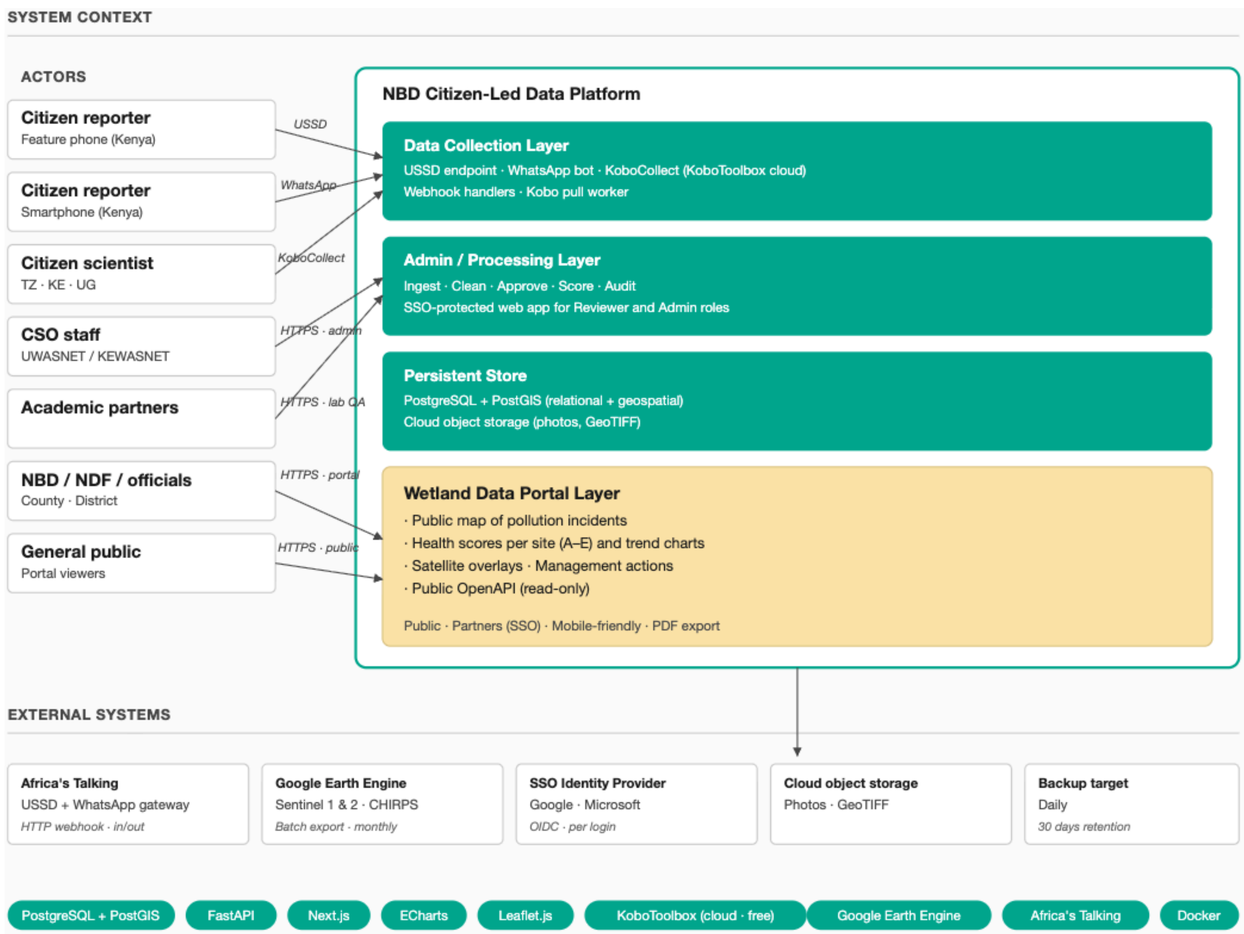

Figure 1 — System Context. All actors, system boundaries, data flows

Guiding Principles

Open source first. All custom components are built under open-source licences. KoboToolbox and all backend components are fully open source. Google Earth Engine, used for satellite data processing, has a generous free tier and a long-standing commitment to support non-commercial use.

Low-connectivity resilience. USSD runs on the GSM voice network only. KoboCollect captures data offline and syncs on reconnect.

Community data ownership. Data collected by citizens belongs to the communities and partner organisations. The platform is designed to facilitate handover to NBD without vendor lock-in.

Interoperability. Every monitoring site has a persistent, structured identifier (e.g. `NBD-MARA-001`). This identifier is embedded in the KoboCollect form, every admin layer record, every GEE pipeline output, every lab QA record, every FGD session record, and every portal API response. Any external dataset can join to platform data using this key alone — no custom field mapping is needed. Spatial layers are published as GeoJSON via documented API endpoints.

Scope

- Pollution episode reporting via USSD and WhatsApp (Kenya)
- Monthly structured water quality sampling via KoboToolbox (Tanzania — Mara wetlands; Kenya and Uganda — Sio-Siteko)
- Admin layer for data cleaning and approval
- Wetland data portal with pollution incident map, health scores, and management actions
- Sentinel 1 & 2 / CHIRPS Earth Observation data integration via Google Earth Engine
- Laboratory QA data ingestion
- FGD session data capture for indigenous knowledge

3.1 Constraints & Hardware Requirements

User-side Devices and Roles

Role	Device & OS minimum	Connectivity	Location

Citizen reporter — USSD (Kenya)	Any GSM handset	GSM voice only; no data required	No device GPS. User picks sub-county from a menu; platform geo-codes it server-side.
Citizen reporter — WhatsApp (Kenya)	Smartphone; Android 7.0+ / iOS 12+	Intermittent mobile data	User picks sub-county from a menu. platform geo-codes it server-side.
Citizen scientist (TZ UG, KE)	Android 7.0+; KoboCollect v2024.x; ≥ 2GB free storage	Intermittent data; offline-capable	Device GPS; form requires ≤ 20 m accuracy
CSO staff / Academic partner	Laptop, tablet, or smartphone; any current browser (last 2 major versions)	Medium-to-low bandwidth	n/a
NBD / NDF / officials / public	Any browser-capable device; (last 2 major versions)	Reliable internet	n/a

KoboCollect stores survey data on the device when there is no connectivity. It syncs automatically to the KoboToolbox server when the device reconnects.

Citizen scientists and wetland watchers are trained through a Train-the-Trainers (ToT) model. The platform must be fully usable from the first day of deployment by users who received training second-hand, without requiring IT support.

Handheld multi-parameter probes,used by citizen scientists to measure pH, temperature, and dissolved oxygen,are provided by the project. These are separate hardware items. The data they produce is entered through the KoboCollect form.

Hosting Infrastructure

- Hosting model: **Cloud-based, Dockerised**
- Production server: **8 GB RAM, 4-core CPU, 30 GB storage**
- Media attachments (photos, documents) stored in cloud object storage, separate from the VM

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4. Architecture Design

4.1 System Layers

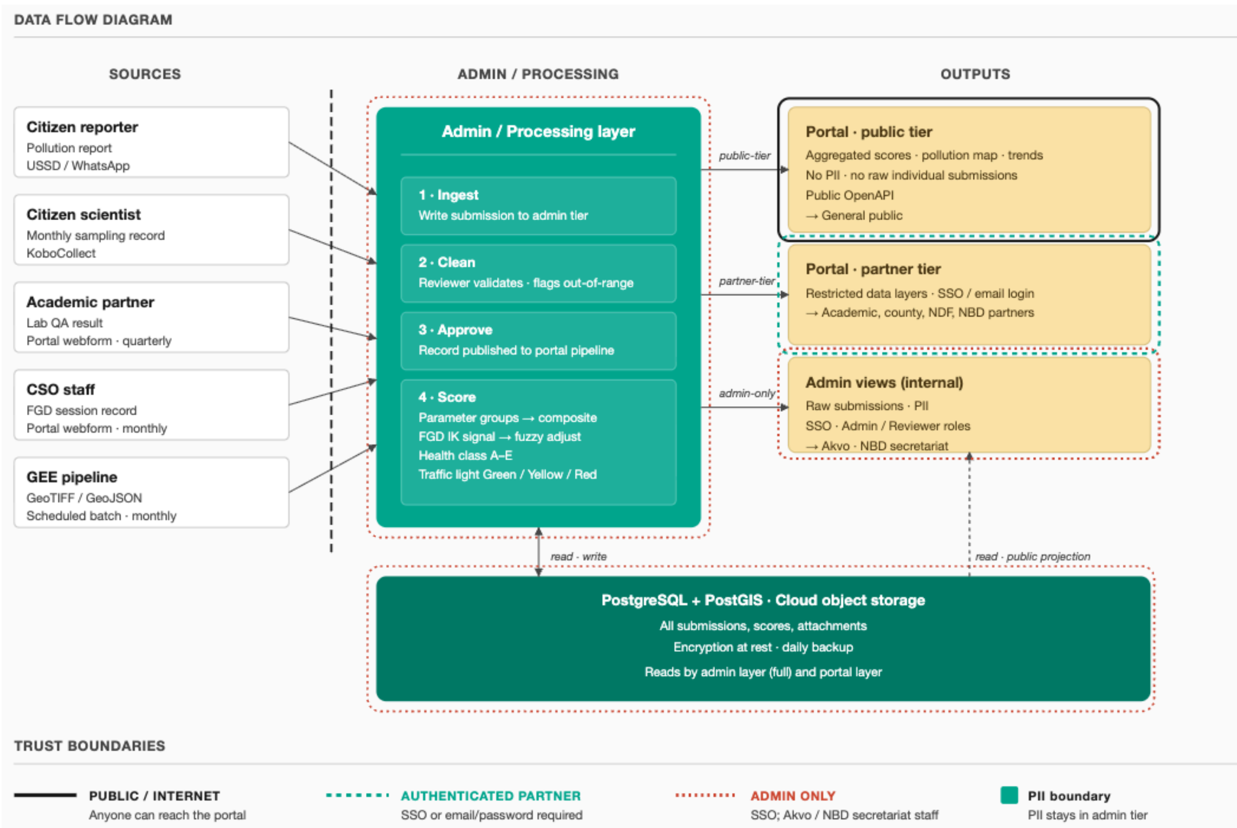
The platform has three layers. The two data collection workflows maintain separate pipelines through the admin layer, then converge at the output stage.

Data Collection Layer — the citizen-facing interfaces. Wetland watchers submit pollution episodes via USSD or WhatsApp Business bot. Citizen scientists submit monthly structured sampling data via KoboCollect.

Admin / Processing Layer — the internal layer where staff clean and approve submissions. The pollution and sampling pipelines remain separate. Scoring is automated from approved data. External data (Sentinel 1 & 2, CHIRPS, lab QA results) and FGD session records are also ingested here.

Wetland Data Portal Layer — the public-facing output. The two pipelines converge in a single portal showing pollution incidents on an interactive map, wetland health scores, management actions, and satellite overlays. All views are accessible on mobile and can be exported as PDFs.

The diagram below shows how all data sources flow through the admin layer to reach the portal. The trust boundaries — public, partner, and admin-only — are annotated.



4.2 Pollution Episode Reporting (Kenya)

Actors: wetland watchers — community members who report pollution events as they occur.

Incident types reported: unusual water colour, smell, fish or other animal kills, changes in water levels.

Two parallel channels serve the same reporting flow:

Channel A — USSD (Feature Phone Users)

Feature phone users dial a dedicated short code and navigate a branching menu.

```
Wetland watcher dials short code
          │
          ▼
USSD branching menu presented
          │
          ├── Select incident type
          │            │
          ├── Select location
          │
          ▼
Report submitted to admin layer
```

The flow runs entirely over the GSM voice network. No mobile data, smartphone, or app installation required.

Channel B — WhatsApp Business Bot (Smartphone Users)

```
Wetland watcher opens WhatsApp
          │
          ▼
Menu-driven bot flow initiated
          │
          ├── Select incident type
          │
          ├── Select sub-county
          │
          └── Attach GPS-stamped photo or voice note (optional)
          │
          ▼
Report submitted to admin layer
```

WhatsApp users can attach photos and voice notes alongside their submission. Both channels use Africa's Talking as the telco gateway and feed into the same admin-layer pipeline.

Admin Layer — Pollution Pipeline

1. **Ingest** — report received from USSD or WhatsApp webhook; written to the admin layer
2. **Clean** — staff review for completeness; remove obviously erroneous entries
3. **Approve** — submission approved for publication to the wetland data portal
4. **Traffic light classification** — approved records contribute to basin health status (Green / Yellow / Red)

4.3 Monthly Structured Water Quality Sampling (Tanzania-Mara, Kenya/Uganda-Sio-Siteko)

Actors: citizen scientists — trained community volunteers who visit designated sampling sites once per month.

Collection tool: KoboToolbox, accessed via the KoboCollect Android app. KoboToolbox is open source and has robust offline functionality.

All numeric fields in the KoboCollect form have min-max constraints configured at the form level. This catches obvious probe entry errors before the form is saved. Out-of-range entries prompt the citizen scientist to re-enter the value before the form can be submitted.

Data Collection Flow

```
Citizen scientist arrives at sampling site (e.g. NBD-MARA-001)
        │
        ▼
Opens KoboCollect on Android smartphone (offline mode)
        │
        ▼
Completes geo-referenced structured form:
  ├── Physico-chemical measurements
  │
  ├── Ecological observations
  │
  ├── Hydrological observations
  │
  └── GPS-stamped photographs
        │
        ▼
Form saved locally on device (no connectivity required)
        │
        ▼
Connectivity available → KoboCollect syncs to KoboToolbox server
        │
        ▼
Submission received by KoboToolbox         │
        ▼
KoboToolbox API → processing service pulls new submissions
        │
        ▼
Admin layer — sampling pipeline
```

Admin Layer — Sampling Pipeline

Same steps as the pollution pipeline: Ingest → Clean → Approve. After approval, the scoring engine computes parameter group scores and the composite score automatically.

Shadow Sampling and Lab QA Integration

Every quarter, academic partners (Makerere University / University of Nairobi) conduct shadow sampling at the same sites as citizen scientists. They collect independent samples and test for all citizen scientist parameters, plus Biochemical Oxygen Demand, Orthophosphate, Nitrate, and Mercury.

Shadow sampling results are compared against citizen scientist submissions to validate measurements and flag systematic discrepancies for retraining.

Lab QA results are ingested as a separate record type in the admin layer, linked to the site identifier and sampling period.

4.4 Admin Interface

The admin interface is the internal processing layer. It is a web application accessible to authorised staff only,

Roles

Two roles are defined. Admin is a strict superset of Reviewer.

Role	Permissions
Reviewer	Review and approve or reject pollution reports, sampling records, lab QA reports, and FGD session records
Admin	All Reviewer permissions plus: invite users and assign roles; create and manage monitoring sites; delete records

Sidebar Navigation

Both roles see **Data** as the primary workspace item. Admins additionally see **User management** and **Site management** under a Management section. Reviewer accounts see these items but they are locked.

Data Screen

The Data screen is the central working view. All submitted records appear in a single filterable list. Three selectors narrow the list:

- **Form** — Pollution report · Sampling data · Lab QA report · FGD session
- **Status** — Pending approval · Approved
- **Basin** — Mara Basin · Sio-Siteko

The record count updates live as selectors change. A **Clear** control resets all three selectors.

Each row represents one submission. A Form chip (colour-coded by type) and a Basin / Site chip identify the record at a glance. Clicking a row expands it inline to show the full submission detail. The row collapses on a second click.

Actions available in the expanded view:

Status	Reviewer actions	Admin additional actions
Pending approval	Approve · Reject · Edit	Delete
Approved	View	Edit · Delete

For sampling data records, the expanded view shows the automated score and health class (A–E). These values are read-only — they are computed from submitted measurements.

Add New

A blue **+ Add New** button sits right-aligned in the filter bar. Clicking it opens a modal to select a form type and basin, then launches the corresponding webform.

Four form types are available:

Pollution report — basin-level; no specific site required.

Sampling data — the modal additionally prompts for the specific monitoring site.

Lab QA report — entry point for academic partners entering lab results. Links to the relevant sampling record and site.

FGD session — entry point for CSO staff entering structured indigenous knowledge data from Monthly Baraza sessions. The form captures: - Date of Baraza - Basin and associated sampling site - Number of FGD participants - Structured responses per dimension (dropdown): - Fish abundance change: Same or increased / Slightly declined / Moderately declined / Severely declined - Water clarity change: Same or clearer / Somewhat worse / Much worse - Vegetation cover change: Same or more / Partially lost / Severely lost - Pollution events reported: None / Occasional / Frequent - Open notes (stored but not scored) - Facilitator name

FGD records link to a basin and sampling period. The scoring pipeline reads the most recent FGD record for each site and period to compute the indigenous knowledge signal used in the fuzzy logic adjustment (see Section 5.6).

```
Monthly Baraza (FGD session)
          │
CSO staff enters FGD session record via Add New
          │
          ▼
Admin layer — FGD record
  site:          NBD-MARA-001
  period:        June 2026
  fish_decline:  moderately_declined  → 0.6
  clarity:       much_worse           → 1.0
  vegetation:    partially_lost       → 0.4
```

```
       |
       ▼
Scoring pipeline reads record
→ IK signal = average(0.6, 1.0, 0.4) = 0.67
→ fuzzy adjustment applied to composite score
```

Data added via Add New webforms is moved directly to Approved status.

User Management (Admin only)

Displays all platform users as cards showing name, email, organisation, and assigned role. Admins can invite new users by email and assign Admin or Reviewer roles. Users cannot self-register.

Authentication: the admin interface uses SSO via common identity providers (Google, Microsoft). No passwords are managed by the platform.

Site Management (Admin only)

Lists all registered monitoring sites in a table with columns for Site ID, name, country, basin, and coordinates. Admins can add new sites, edit site metadata, or disable a site (which hides it from the Add New form but retains its historical records). Site identifiers follow the format `NBD-MARA-001`.

Each site record supports attached documents. Supported formats: PDF, JPG, PNG (maximum 10 MB per file). Documents are stored in cloud object storage, linked to the site by Site ID. Attachments are visible to Admins and Reviewers in the expanded site view. They are not publicly visible on the portal.

Examples of attached documents: wetland management plans, baseline assessment reports, field notes, boundary maps.

4.5 Wetland Data Portal

The portal gives NBD Secretariat, NDFs, county and district officials, CSOs, academic partners, and communities access to collected data, derived health scores, and management actions.

Key features: - **Pollution incident map** — geospatial display of reported pollution episodes by basin - **Wetland health scores** per site, per time period - Trend visualisation over time - Satellite data overlays (vegetation indices, water surface extent, land use change) - Mobile-friendly layout; exportable as PDF for offline use at Barazas and government meetings

Data sources: pollution episode outputs + monthly sampling outputs + external data (Section 4.7)

Access: - Public view: health scores, pollution incident map, site-level detail, trend data — no login required - Partner view: restricted data layers — SSO or email and password login; no self-registration; accounts created by Admin

Data refresh: weekly

Tech stack: Next.js (frontend), Leaflet.js (maps), ECharts (charts), FastAPI backend API, PostgreSQL database

4.5.1 Accessibility & Compliance

The portal targets WCAG 2.1 Level AA conformance.

- All map content has a parallel text or table view, so a screen-reader user reaches the same information without using the map.
- Colour contrast is ≥ 4.5:1 for body text and ≥ 3:1 for the traffic-light status chips. Traffic-light status is always paired with a text label, never colour-only.
- All interactive controls are reachable and operable by keyboard. The focus outline is preserved (not removed by CSS reset).
- Semantic HTML headings (h1–h3) in document order; page lang set; every non-decorative image has alt.

4.5.2 Performance

Text content — health scores, incident lists, trend charts — loads quickly (< 5s) on low-bandwidth connections. Map tile and satellite raster overlay load times depend on connection quality and are not subject to a fixed target; rasters are served at display resolution to keep file sizes as small as possible.

4.5.3 Search and Filter

Users can narrow the portal view by:

- Basin — Mara · Sio-Siteko
- Site — any of the eight NBD-<BASIN>-<NNN> sites
- Date range — preset (last 30 days, last quarter, last year) or custom
- Health class — A · B · C · D · E (multi-select)
- Pollution incident type — water colour change, smell, fish or animal kill, water-level change (multi-select)
- Free-text search — across site name and description

Filter state is serialised into the URL query string so any view can be shared by link or bookmarked.

4.5.4 Public API Strategy

The FastAPI backend auto-generates an OpenAPI specification. A documented subset(will be updated in LLD if changed) of read-only endpoints is exposed publicly under a stable contract:

- GET /api/v1/sites — list all sites with metadata and current health class.
- GET /api/v1/sites/{site_id} — site detail including geometry.
- GET /api/v1/sites/{site_id}/scores — historical health scores and traffic-light status.
- GET /api/v1/sites/{site_id}/external/{source} — external data joined to the site (Sentinel NDVI, CHIRPS rainfall, etc.).
- GET /api/v1/incidents — pollution incidents, aggregated, no PII.

Partner endpoints (raw submissions, individual reporter contact, full audit log) sit behind SSO. They are not part of the public contract.

The platform publishes:

- OpenAPI spec ,human-readable docs at /api/docs.
- Data dictionary documenting every field returned and its units.
- Versioning policy: all URLs prefixed /api/v1/; the v1 contract is frozen at go-live; breaking changes ship under /api/v2/, with v1 supported for at least 12 months after v2 release.
- Fair-use rate limit: 60 requests / minute / IP on public endpoints (HTTP 429 on excess); higher limits available to partners on request.

4.6 Technology Stack

None of the components below are experimental. They are the tools the team uses on citizen-data and environmental reporting platforms in this region, and the selection reflects both that experience and the constraints the requirements gathering and field reconnaissance surfaced: intermittent connectivity at monitoring sites, feature-phone access, limited digital literacy among citizen reporters, a tight cost ceiling, and a handover target of NBD staff operating the platform independently after the pilot. Each decision record below names the alternatives considered and why they were set aside.

4.6.1 Backend API — FastAPI (Python)

The geospatial workload is already Python — GeoAlchemy2, Shapely, and rasterio for GEE result ingestion — so a Python backend keeps the stack consistent. Django REST Framework was considered and rejected: its conventions and bundled components suit larger, multi-domain applications, and this portal's scope does not justify that overhead.

FastAPI is lighter, auto-generates the OpenAPI spec, and matches the team's established pattern for portals of this complexity. Python skills are widely available across the Nile Basin partner pool. The async-first style takes ramp-up for newcomers, but the community is mainstream and growing. The OpenAPI contract is the abstraction that makes the backend replaceable: re-implementing in Django + DRF or Flask + APISpec leaves the portal and mobile channels unchanged.

4.6.2 Frontend (Portal + Admin) — Next.js

Next.js provides mature server-side rendering, good portal SEO, and fast mobile first-paint. A shared component library covers both the public portal and the admin interface. React skills are the most widely available frontend profile in East Africa, which matters for long-term maintainability. The frontend consumes the FastAPI REST API and holds no application logic, so if NBD needs to move away from Next.js later, the API contract is unchanged and the swap is a frontend-only exercise. The main risk is Vercel-driven release pace between major versions; we pin and upgrade on a deliberate release cadence. The platform is self-hosted in our Docker stack, not on Vercel, so there is no commercial dependency on Vercel services.

4.6.3 Database — PostgreSQL + PostGIS

PostGIS is the de-facto standard for geospatial relational work — mature, supported by every major cloud, with deep tooling. MySQL with spatial extensions was considered but offers a thinner geospatial layer. MongoDB with geospatial indexes was considered but a relational model fits the structured tabular data better. Postgres skills are abundant; PostGIS training is widely available. At pilot scale, tuning complexity is not a concern. At scale-up, managed Postgres services on any major cloud run at roughly $50–150 per month.

4.6.4 Satellite Processing — Google Earth Engine

The free non-commercial tier covers the pilot volume, and GEE provides ready-access to pre-computed Sentinel and CHIRPS catalogues with no raster-processing infrastructure to operate or maintain. Alternatives considered were Sentinel Hub (commercial, ~$500–2,000 per month for equivalent compute), a self-hosted Sentinel pipeline. None match GEE's cost profile at pilot scale.

Two risks are documented. First, licence tier: if NBD later monetises derived data, the non-commercial terms would require an upgrade to the GEE commercial licence. Second, discontinuation: low probability given Google's long-running public commitment, but non-zero. Both risks are mitigated by the architecture: every GEE script is exportable, and derived index values are stored in PostgreSQL, so historical results survive a tier transition regardless of what happens to GEE. The documented migration path is creating our own pipelines which is disproportionate to the pilot needs and the use of satellite data in the platform

4.6.5 USSD & WhatsApp Gateway — Africa's Talking

No other provider offers equivalent USSD and WhatsApp Business coverage in East Africa under a single API contract. Twilio has no USSD in Kenya. Infobip's regional coverage and support quality are weaker. Hover SDK delivers USSD via an Android app rather than the network, which is unusable on feature phones. Africa's Talking is East Africa-based and the dominant operator for exactly this use case. The primary risk is vendor lock-in on short-code allocation — short codes are tied to the AT account — and cost growth with submission volume. At pilot scale, cost is roughly $50–150 per month. If AT were discontinued, USSD could be sourced directly through a telco aggregator (slow to procure) and WhatsApp via the Meta Cloud API, now generally available.

4.6.6 Data Collection — KoboToolbox (cloud, free tier)

KoboToolbox on the public cloud (kf.kobotoolbox.org) was chosen over self-hosting. Self-hosting was considered and rejected for the pilot: it adds container surface area not needed at pilot volume, and the free tier comfortably covers the expected submission and storage rate (~32 sampling submissions per month at ≤ 5 MB each). Alternatives considered were ODK Central (SaaS or self-hosted), SurveyCTO, and raw ODK. KoboToolbox is open-source, has strong offline behaviour through the KoboCollect Android app, and its UX is already familiar to many partner organisations in the region. No infrastructure for Akvo or NBD to maintain. The risks are the usual dependency on cloud availability and annual re-checking of free-tier limits against actual volume. If either limit were hit, self-hosting KoboToolbox on the same cluster is a documented path, and ODK Central is a second drop-in replacement sharing the XLSForm spec.

4.6.7 Visualisation — ECharts (charts) and Leaflet.js (maps)

All mature charting libraries produce equivalent chart output; they differ in component API, configuration style, and default visual language. Swapping one for another is a frontend-only change with no impact on data or the API. ECharts is the team's working preference and the one the team delivers fastest in. Chart.js and Plotly were the alternatives considered. Leaflet is the lightest mature mapping option for our polygon and marker layers, with no proprietary tile dependency. Nothing in the portal's requirements points to a reason to use Mapbox GL or OpenLayers — both would add complexity without adding capability at this scope. Tile hosting costs are negligible via OSM-compatible providers. Both libraries are fully open-source.

4.6.8 Infrastructure — Docker

Containerised services are portable between cloud environments and give a clean, reproducible deployment. Raw VM with systemd units was considered but offers no portability. Kubernetes was considered but imposes significant operational overhead not justified at pilot scale. Docker Compose is the right tool for a four-container application. If volume at scale-up requires Kubernetes, Compose-to-Kubernetes translation is a straightforward one-week task.

4.6.9 Codebase & CI/CD — GitHub + GitHub Actions

GitHub provides the lowest friction for a multi-organisation contributor base, built-in CI, and mature dependency scanning. GitLab (SaaS or self-hosted) and a self-hosted Gitea + Jenkins combination were considered; both add operational overhead. Vendor lock-in is low — git history and Actions YAML are portable and the YAML conversion to GitLab CI is well documented.

All components above are open-source licensed, except Africa's Talking (commercial SaaS) and Google Earth Engine (free non-commercial tier).

The cost profile is deliberate. A citizen data platform should be cheap to run, particularly in its first phase — cost is a sustainability constraint, not a secondary consideration. Every component was chosen with an eye to minimising the recurring bill and keeping the platform operable without Akvo after handover.

Replacing GEE or KoboToolbox with self-hosted alternatives is possible, but both would add infrastructure weight — a raster-processing pipeline or a managed form service — without improving Phase 1 outcomes. Those paths are documented as fallbacks and deferred to a later phase if the free-tier terms change or the volume outgrows them.

NBD and its national partners can operate and maintain the platform independently after the pilot.All components are open-source licensed except the Africa’s Talking API (commercial SaaS) and Google Earth Engine (free tier, non-commercial use).

The platform is designed so that NBD and its national partners can operate and maintain it independently after the pilot period.

4.7 External Data Integration

Purpose: augment citizen-collected ground-truth data with authoritative external datasets.

Known sources — Phase 1:

- **Sentinel 1 & 2** (Earth Observation) — vegetation indices, water surface extent, land use change
- **CHIRPS** — precipitation and climate data
- **Lab QA results** from academic partners — heavy metals, nutrients, biochemical oxygen demand, orthophosphate, nitrate, mercury

Sentinel 1 & 2 imagery and CHIRPS precipitation data are processed using Google Earth Engine (GEE). GEE scripts extract vegetation indices, water surface extent, and land use change indicators and export results as GeoJSON or GeoTIFF for ingestion into the platform database.

Anticipated future sources: - Data from Earthwatch - Additional EO datasets

Integration pattern: - Sentinel 1 & 2 / CHIRPS: automated batch pipeline (scheduled jobs) pulling and processing data via GEE; results ingested as structured records - Lab QA results: submitted by academic partners via the portal webform (Add New → Lab QA report) - Format standards: GeoTIFF (EO data), CSV (lab results), REST API (future sources) - Temporal alignment: monthly

Data provenance: every external dataset record carries source name, collection date, and version. This is displayed alongside derived scores in the portal.

4.7.1 Data Licensing and Downstream-Use Matrix

Source	Licence	Permits commercial / monetised re-use?	Attribution required?	Implication for NBD if/when monetising portal data
Sentinel 1 & 2 (Copernicus)	Free, full and open	Yes — explicitly	Yes — attribute Copernicus / ESA	None — safe for commercial derivative products

CHIRPS (UCSB Climate Hazards Center)	Public domain	Yes	Citation recommended	None
Google Earth Engine (compute service)	Non-commercial tier free; commercial use requires GEE Commercial licence	No under the free tier	N/A	Binding constraint. NBD must upgrade to the GEE Commercial tier, or migrate the satellite-processing to a custom pipeline with its own compute engine

4.7.2 Integration Resilience and Reconciliation

Each external integration has a documented failure-handling contract.

- Africa's Talking webhook (USSD + WhatsApp). Delivery is at-least-once. Every report carries an idempotency key, so retries do not double-count. The webhook receiver returns HTTP 200 only after the message is durably persisted. AT retries failed deliveries with exponential backoff on its side. An alert fires if the webhook receiver is unavailable for > 5 minutes.
- KoboCollect → KoboToolbox → admin pull. KoboCollect retries indefinitely on the device. The admin-layer pull worker runs every 60 minutes, uses a watermark cursor on submission_time, and writes to a dead-letter table on schema mismatch. The reviewer triages dead-letter entries.
- GEE batch. Scheduled monthly. Processes Sentinel and CHIRPS data in GEE and writes derived index values to PostgreSQL as aggregated rows. Where a map overlay tile is generated, it is stored in Cloud Storage only if the value has

changed from the prior month's export. On failure, re-run automatically up to 3 times with 1-hour backoff. Outage policy. The portal continues to serve last-known-good data with a visible "as of" timestamp. No external-integration outage takes the portal itself down.

4.8 Non-functional Requirements

Targets are sized for the pilot workload: two basins, eight sampling sites, approximately 20 active citizen scientists, and approximately 50 citizen reporters. They will be re-baselined against real telemetry at the end of Year 1.

Availability. Platform availability target is 99% per month (excludes pre-announced maintenance windows). The USSD and WhatsApp pipeline availability is bounded by Africa's Talking SLA.

Performance. Portal pages load within 2 seconds on a 3G connection. API responses for cached site and score data return within 800 ms. A pollution report submitted via USSD or WhatsApp is visible in the admin interface within 30 seconds. A KoboCollect submission is visible within 15 minutes of device reconnection.

Concurrent users. The portal is designed for up to 200 simultaneous visitors, with capacity to handle short bursts beyond that through autoscaling. The admin interface serves a small team of up to 10 reviewers and administrators.

Recovery. Recovery Point Objective (RPO): 24 hours — in the event of a failure, at most one day of data may need to be re-entered. Recovery Time Objective (RTO): 4 hours — the platform is restored from backup and redeployed from the GitHub repository within that window. Backups run daily and are retained for 30 days.

Localisation. The interface is in English at MVP. The form and USSD menu structure supports translation; Swahili and Luganda are planned for a post-MVP release.

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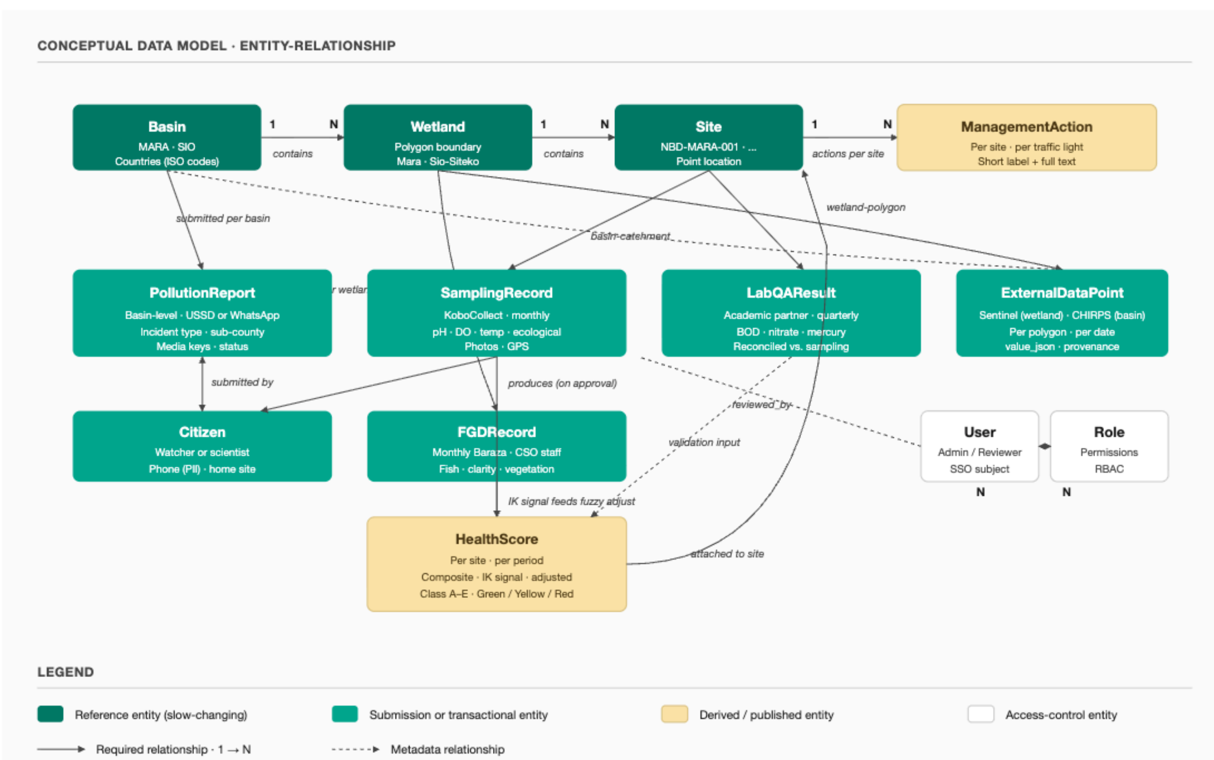
5. Data Model

5.1 Parameters Collected

Category	Parameters	Collected by
Physico-chemical	pH, Temperature, Dissolved Oxygen	Citizen scientists (handheld multi-parameter probes)
Ecological	Fish Catch Per Unit Effort (CPUE)	Citizen scientists

Category	Parameters	Collected by
Hydrological / catchment	Water levels/flow, Flooding extent	Citizen scientists + wetland watchers
Pollution episodes	water colour, smell, fish or other animal kills, or change in water levels	Wetland watchers (USSD/WhatsApp)
Lab validation	Biochemical Oxygen Demand, Orthophosphate, Nitrate, Mercury, Heavy metals and nutrient loads (N/P)	Academic partners (shadow sampling)
Indigenous knowledge	Community observations captured through FGD sessions at Monthly Barazas	CSO staff (via admin FGD session form)

5.2 Conceptual Data Model



The platform organises data across three levels of geographic hierarchy: Basin, Wetland, and Site. Every data record attaches to one of these three levels, and that attachment determines which identifier it carries.

The basin is the top-level container. Two basins are configured for Phase 1: Mara (Kenya/Tanzania) and Sio-Siteko (Kenya/Uganda). Pollution reports attach at basin level — citizen reporters select a sub-county, not a specific monitoring site. Basin-catchment external data (CHIRPS rainfall) also attaches here.

Wetland sits within a Basin. FGD session records attach at wetland level — a Monthly Baraza is a wetland-community meeting, not a site-specific event. Wetland-polygon external data (Sentinel 1 & 2 indices) also attaches here, because Sentinel indices are computed over the wetland boundary polygon, not a point.

Site (e.g. NBD-MARA-001) is the most specific level — a fixed point location visited monthly by citizen scientists. Sampling records, lab QA results, health scores, and management actions all attach to a site.

The main data entities are:

PollutionReport — submitted by a Citizen (watcher role) via USSD or WhatsApp at basin level; records incident type, sub-county, and optional media.

SamplingRecord — submitted by a Citizen (scientist role) via KoboCollect at site level monthly; records physico-chemical, ecological, and hydrological measurements plus GPS location and photos.

LabQAResult — submitted by an academic partner quarterly at site level via the admin portal; records lab parameters (BOD, nitrate, mercury, orthophosphate, nutrient loads).

FGDRecord — submitted by CSO staff at wetland level monthly; captures structured indigenous knowledge from Baraza sessions across three dimensions: fish abundance, water clarity, and vegetation cover.

ExternalDataPoint — ingested on a monthly schedule; attaches at wetland-polygon scope (Sentinel 1 & 2) or basin-catchment scope (CHIRPS).

HealthScore — a derived entity computed per site per period from an approved SamplingRecord, adjusted by the FGD indigenous knowledge signal through the fuzzy logic pipeline (see §5.6). It does not exist until a sampling record is approved and scored.

ManagementAction — editorial content maintained by the NBD Secretariat; defined per site and per traffic light status; displayed publicly on the portal.

Citizen and User are distinct. Citizens (watchers and scientists) are data submitters, identified by phone number (PII). Users are admin-layer staff (Admin and Reviewer roles), authenticated via SSO.

Three interoperability keys tie the model together: site_id for sampling records and lab QA; wetland_id for FGD records and Sentinel data; basin_id for pollution reports and CHIRPS data. Any external dataset can join to platform data using the appropriate key without custom field mapping.

5.3 Logical and Physical Data Models

Detailed schemas — attribute definitions, data types, validation constraints, and the PostgreSQL + PostGIS DDL — will be developed as part of the platform build and documented in a Low-Level Design (LLD). The LLD will be shared with NBD in due course. This document limits itself to high-level design.

They will be available at the github repository . Keeping them in the same place as the code will also help with the maintainability of the schemas in case of updates

5.4 Site Identifiers

Persistent, structured site identifiers follow the format `NBD-MARA-001`, `NBD-SIO-001`, etc. Four sampling sites per wetland. The site ID must be consistent across all data layers.

The site ID is the primary key for interoperability. It is embedded in every layer of the platform:

```
KoboCollect form      →  site_id field (fixed, pre-loaded per device)
          │
          ▼
Admin layer record    →  site_id: NBD-MARA-001
          │
          ├──▶ GEE pipeline output    {"site_id": "NBD-MARA-001", "ndvi":
0.72}
          │
          ├──▶ Lab QA record          site_id: NBD-MARA-001, period: 2026-06
          │
          ├──▶ FGD session record     site_id: NBD-MARA-001, period: 2026-06
          │
          └──▶ Portal API response    GET /api/sites/NBD-MARA-001/scores
```

Any external dataset can join to platform data using the site ID alone. No custom field mapping is needed.

5.5 Wetland Health Scores

Health scores are calculated at the parameter level, aggregated into group means, and combined into a single composite score per site.

Health Classes (upper bound inclusive)

Class	Label	Score range	Description
A	Very Good / Natural	0.8 – 1.0	Unmodified or natural; very high ecological integrity
B	Good / Slightly modified	0.6 – 0.8	Largely natural with few modifications; small loss of natural habitat
C	Moderate / Moderately modified	0.4 – 0.6	Moderate change in ecosystem processes and loss of natural habitats
D	Poor / Largely modified	0.2 – 0.4	Large change in ecosystem processes; serious loss of natural habitat and biota
E	Very Poor / Critically modified	0.0 – 0.2	Critical modifications; ecosystem processes completely altered

Parameter Groups and Scoring

Parameter group	Parameters scored
Physico-chemical	Water temperature, pH, Dissolved Oxygen (converted to Water Quality Index score)
Catchment and hydrological	% wetland converted to non-wetland use; r
Ecological	% wetland area covered by invasive macrophytes; Fish Catch Per Unit Effort;

Each group is scored on a 0.0–1.0 scale. Group scores are averaged to produce the composite score.

Scoring Pipeline

```
Monthly sampling record (approved)
        │
        ▼
┌──────────────────────────────────────────────────┐
│  Score each parameter group (0.0–1.0)            │
│                                                  │
│  Physico-chemical  ─────────────▶ 0.68           │
│  Catchment/hydro   ─────────────▶ 0.58           │
│  Ecological        ─────────────▶ 0.65           │
│                                                  │
└──────────────────────────────────────────────────┘
                    │
                    ▼
         Composite = average = 0.638
                    │
     ┌──────────────────────────────────┐
     │  FGD session IK signal           │
     │  fish: moderately declined       │
     │  clarity: much worse             │
     │  vegetation: partially lost      │
     │  encoded signal = 0.67           │
     └──────────────────────────────────┘
                    │
                    ▼
       Fuzzy logic adjustment
       0.638 → 0.55
                    │
                    ▼
     ┌──────────────────────────────────┐
     │  Health class: C (Moderate)      │
     │  Traffic light: YELLOW           │
     │  Action: community response      │
     └──────────────────────────────────┘
```

Fuzzy Logic Adjustment

A composite score of 0.638 sits on the border between Yellow and Green. The number alone does not capture whether the wetland is degrading relative to its historical state. Indigenous knowledge from FGD sessions at Monthly Barazas provides that context as a soft signal.

Step 1 — Encode the IK signal.

FGD session records use structured dropdowns per dimension. Each response maps to a numeric score. The scores are averaged into a single IK signal (0.0 = no decline reported; 1.0 = severe decline reported).

Dimension	FGD response	Encoded score
Fish abundance	Moderately declined	0.6
Water clarity	Much worse	1.0
Vegetation cover	Partially lost	0.4
IK signal		**0.67**

Step 2 — Express inputs as fuzzy sets.

Both inputs (composite score and IK signal) are assigned membership values across three fuzzy sets: low, medium, high. A score of 0.638 falls mainly in the "medium" set. An IK signal of 0.67 falls mainly in the "moderate" set.

Step 3 — Apply if-then rules.

Rules fire based on the combination of input fuzzy sets:

Composite score	IK signal	Output
High	None	High
High	Moderate	Medium
High	Strong	Medium
Medium	None	Medium
Medium	**Moderate**	**Low** ← fires
Medium	Strong	Low
Low	Any	Low

Step 4 — Defuzzify.

The fired rule output is converted back to a single number using the centroid method.

Result: composite 0.638 → adjusted score **0.55** → traffic light **YELLOW** → community response triggered.

Without the FGD data, 0.638 would classify as Green. No action would be taken.

Traffic Light Classification

Status	Score threshold
Green	*> 0.6*
Yellow	*0.4 - 0.6*
Red	*0 - 0.4*

Management Actions

Management actions are displayed publicly on the portal alongside the traffic light status for each site. Each action shows a short 3-word label as the primary display, with the full description available on expansion. Actions are defined per wetland by the NBD Secretariat and national partners and updated via the admin interface.

Green — No action required. The wetland is within acceptable health bounds. Monitoring continues at the regular monthly cadence.

Yellow

- Establishment of Silt Traps and Grass Strips: Install vegetative filters along the edges of agricultural plots to catch sediment and nutrient runoff (phosphates/nitrates) before they reach the water.
- Constructed Wetlands for Tertiary Treatment: Encourage small-scale, man-made wetlands at the edge of settlement zones to naturally treat domestic greywater.
- Promotion of Livelihoods: Transition farmers in the buffer zone from high-input crops (like sugarcane or maize) to low-impact activities such as apiculture (beekeeping) or sustainable papyrus harvesting.
- Riparian Re-vegetation: Conduct targeted planting of indigenous species (e.g., Ficus sycomorus or Typha) in degraded patches to maintain the 0.4–0.6 ecological integrity score.
- Establishment of Community Conservation Areas (CCAs): Designate no-catch zones or seasonal closures during fish spawning periods to allow stocks to replenish.
- Introduction of Co-Management Groups: Form Beach Management Units (BMUs) that are empowered to self-regulate gear sizes and monitor illegal fishing, reducing the burden on the central Authority.

Red

- Prompt reporting effluent and/or sewage discharge incidences at the local unit of Ministry of Environment and Water.
- Setting up interceptor STPs to direct sewage to sewage treatment plant/combined effluent treatment plant. A plan including the budget and timeline estimates will be presented in the meeting
- Preventing any further encroachment of the wetland via enacting protected area/buffer zone regulation (e.g. increasing buffer widths) as per the Environment Management & Coordination Act. Any violation should be reported immediately to the County Environment Committee.
- Fishermen should switch to better, lower meshed gillnets to increase the catch per unit effort

- The Environment Management Authority should put together a draft management for Wetland for public review by December 2026

The portal shows the 3-word label on the site health card. Clicking it expands the full action text. All Red actions are publicly visible — no login required. The NBD Secretariat updates action text via the admin interface when escalation steps change.

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6. Integration Design

Integration	Direction	Protocol	Data format	Frequency	Failure handling	Cost	Notes
Africa's Talking USSD	Inbound	HTTP webhook	URL-encoded form params; XML response	Per session (event-driven)	AT-side retry with exponential backoff; idempotency key on report; alert if receiver unavailable > 5 min	~$600–1,200	Stateful session
Whatsapp	In/ Out	HTTP webhook	JSON	Per message	Same as USSD; message dropped after 24h per WhatsApp policy	TBD (if NBD can get certified by Meta and get is own	Menu-driven flow

						number)	
KoboToolbox cloud API	Outbound pull	REST / JSON	JSON	Every 60 minutes	Full pull to the staging table and new records updation to the platform database. Email alert to admins	$0 (free tier)	Public KoboToolbox cloud (eu.kobotoolbox.org); free-tier limits reviewed annually
Sentinel 1 & 2 via GEE	Inbound	GEE export	GeoTIFF / GeoJSON	Monthly	3 retries × 1h backoff; alert on persistent failure	$0 (GEE free tier)	Licence-tier risk defined later
CHIRPS via GEE	Inbound	GEE export	GeoJSON	Monthly	Same as Sentinel	$0	Licence-tier risk defined later
Lab QA results	Inbound	Portal webform	Structured form fields	Quarterly	Form-level validation; reviewer queue; auto-reconcile vs. citizen-scientist value	$0	Submitted by academic partners
Cloud object storage (media)	In/Out	Storage compatible API	Binary	Per upload	Multipart upload; optional cross-region replication	~$20/month	

SSO (Google / Microsoft)	Inbound	OIDC	JWT	Per login	Fallback to email + password for portal partners	$0	
Backup target	Outbound	Cloud snapshot	Native	Daily	30 days retention; restore tested quarterly	~$50/month	
Wetland data portal data feed	Internal	FastAPI REST endpoints	JSON	On-demand (cached 1h at edge)	Standard HTTP error semantics; cached fallback	n/a	Admin layer serves data to portal frontend via documented API

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7. Security, Privacy & Data Governance

This is a public citizen data portal with a small PII surface and a small operational team. The controls below are proportionate to that reality.

7.1 Citizen Data Privacy

All personal identifiable information is stripped before data is displayed on the public portal. Only aggregated data is displayed for pollution episodes.

7.2 Data Classification

Tier	Examples	Who can access
Public	Aggregated wetland health scores, pollution incident map, trend charts	Anyone
Private	Wetland watcher / citizen scientist PII, phone numbers,	Admins only

Tier	Examples	Who can access
	individual episode links	

7.3 Access Control Model

Role	Permissions
Wetland watcher	Submit own pollution episode reports only
Citizen scientist	Submit own sampling records via KoboCollect only
Akvo , NBD	Full platform administration
Public	Read public tier only (portal)

Authentication: - Admin interface: SSO via common identity providers (Google, Microsoft). No passwords managed by the platform. - Portal (partner access): email and password login, SSO. - No self-registration. All accounts are created by an Admin via the invite flow. - Public portal views require no login.

7.4 Data Ownership & Sovereignty

Ownership. NBD and the communities from which data is collected are the data owners. Akvo is the data processor — it operates the platform on behalf of NBD .

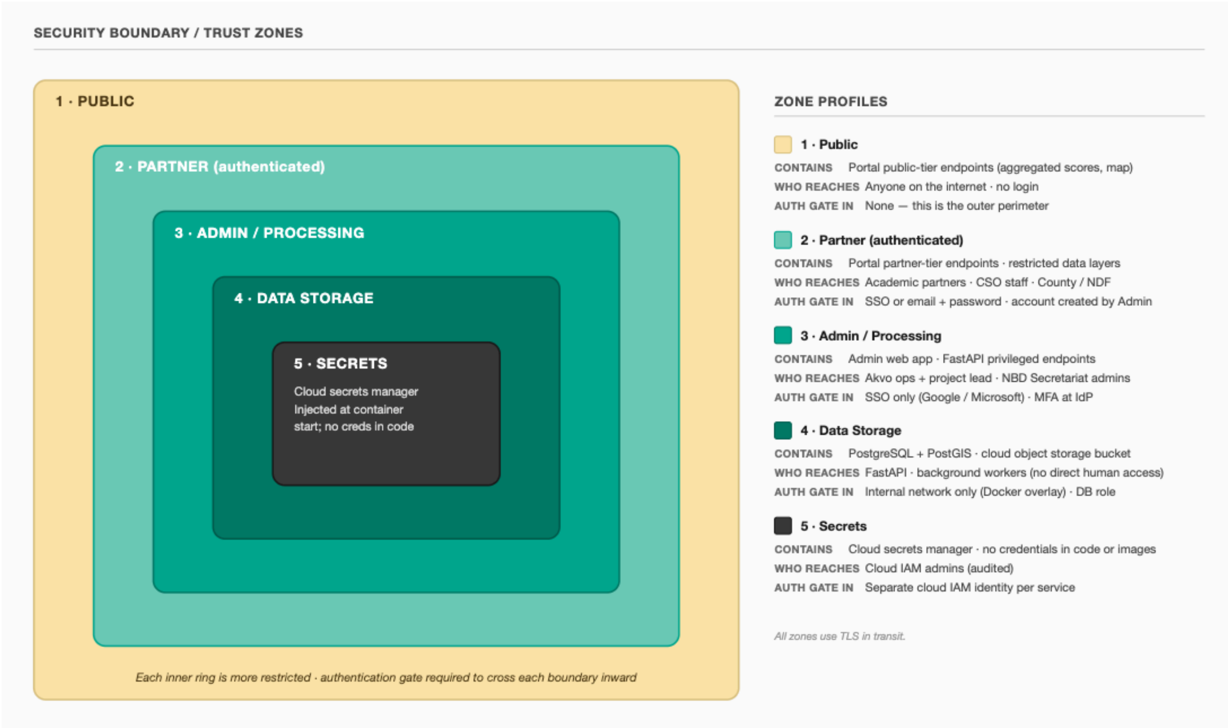
Akvo's role. Akvo has no right to use platform data for any purpose beyond operating the platform for this project. Akvo will not share, sell, or analyse the data for its own purposes.

7.5 Infrastructure Security

- Secrets management: no credentials in code or Docker images; use environment secrets or a secrets manager
- Encryption at rest: all PII and observation data
- Encryption in transit: TLS for all services

7.6 Security Controls

The platform has two trust levels: public and admin. The security design follows from that split.



Public tier. The portal serves aggregated health scores, pollution incidents, and satellite overlays with no login required. No PII appears in any public endpoint — phone numbers are never stored in or returned by the public API. HTTPS is terminated at the GCP load balancer. Rate limiting is applied at the load balancer (configurable, default 60 requests / minute / IP). No additional controls are needed beyond standard GCP network defaults.

Admin tier. The admin interface and FastAPI privileged endpoints are protected by Google SSO. Staff log in with their existing Google accounts — the platform manages no passwords. Two roles are defined: Reviewer and Admin . Every approve, reject, edit, and delete action is written to an audit log with the actor's identity and a timestamp. Accounts cannot be self-created; all invitations go through the Admin role.

PII surface. The only personal data the platform holds is the phone number of each citizen reporter and citizen scientist. Phone numbers are stored in the admin-tier database only. They are never returned by the public API, never included in portal displays, and never exported in any aggregated dataset. Deleting a citizen record removes the phone number while preserving the anonymised submission data.

Secrets. API keys (Africa's Talking, GEE) and database credentials are stored in GCP Secret Manager and injected at container start. No credentials appear in code, Docker images, or files committed to the repository.

Media files. Photos and GeoTIFFs are stored in a private Google Cloud Storage bucket. All access is via signed URLs with a 15-minute TTL. The bucket has no public access policy.

TLS. All traffic — public portal, admin interface, USSD/WhatsApp webhooks, API calls to external services — is encrypted in transit via HTTPS. Internal container-to-container traffic runs on the Docker overlay network and is not externally reachable.

7.7 Incident Handling

7.7 Incident Handling

This is a small platform operated by a small team. The procedure is proportionate.

Platform down or data incorrect. The Akvo operations contact investigates and resolves. GCP monitoring sends email alerts to the operations team on container failures, high error rates, and backup failures. NBD is notified if an outage exceeds 24 hours or if data integrity is affected.

Personal data exposed outside the admin tier. Akvo notifies NBD immediately. NBD notifies the relevant supervisory authority within 72 hours per the applicable data protection law. Affected individuals are informed.

After any significant incident, the Akvo operations team documents what happened and what changed. The note is shared with NBD.

7.8 Data Protection

Citizen reporters and citizen scientists are residents of Kenya, Uganda, and Tanzania. The applicable laws are the Kenya Data Protection Act 2019, the Uganda Data Protection and Privacy Act 2019, and the Tanzania Personal Data Protection Act 2022. NBD is the data controller. Akvo is the data processor. A Data Processing Agreement between NBD and Akvo is signed before the platform goes live.

The platform collects one category of personal data: phone numbers. They are used solely to identify the submitter and send automated acknowledgements. They are not shared with third parties and not used for any purpose beyond operating the platform.

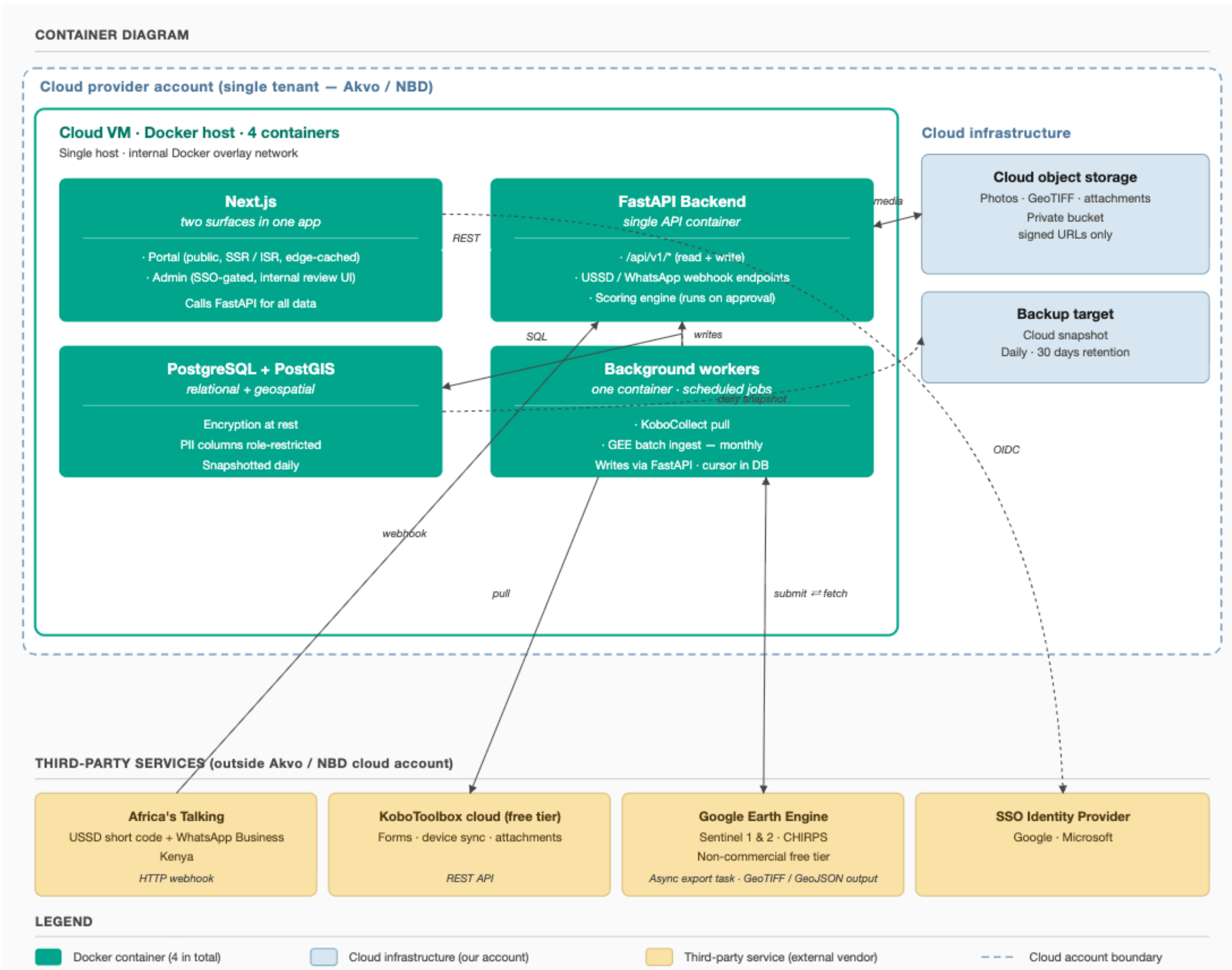
Citizen scientists give written consent at enrolment. Citizen reporters submitting via USSD or WhatsApp receive a one-line consent notice at the start of the menu flow.

GCP has no region in East Africa. Data will be hosted in europe-west1 (Belgium). The Kenyan, Ugandan, and Tanzanian data protection laws permit cross-border transfers where the receiving jurisdiction provides adequate protection. The EU qualifies.

8. Deployment Architecture

8.1 Docker Services

The platform is fully Dockerised. Each major component runs as a separate Docker service.



8.2 Hosting

The platform is cloud-hosted. The architecture also supports local in-country server deployment should NBD choose to operate it independently post-pilot.

Production server specification: 8 GB RAM, 4-core CPU, 30 GB storage. Media attachments (photos, documents) are stored in cloud object storage, separate from the VM. The rasters stored will be optimised to only show changes. This reduced sizing is key to having a reasonably performant platform in resource constrained settings. The primary data will be text and images are already compressed by kobotoolbox and whatsapp at source.

This is a baseline and final resource limits will be set at handover time and adjusted from observed usage.

Three environments are maintained: development, staging, and production.

Codebase. Maintained on GitHub under the Akvo organisation. Continuous integration via GitHub Actions runs automated tests on every pull request. Deployments to staging and production are triggered via GitHub Actions.

8.2.1 Workload Model (Pilot Year 1)

Driver	Value	Source
Active sampling sites	8 (4 Mara, 4 Sio-Siteko)	Inception Report
Active citizen scientists	~20	Inception Report
Active citizen reporters	~50	Inception Report
Sampling submissions / month	~32 (8 sites × ~4 visits)	Monthly cadence per site
Sampling submission size	~400 KB (form + 4 photos at ~100 KB each)	KoboCollect and WhatsApp compress photos at source
Pollution reports / month	~100–300	Estimate; calibrated at end of Y1
Portal monthly visitors	~500–2,000	Estimate; bursty around events

Re-baselined at the end of Year 1 from actual telemetry.

8.2.2 Sizing Rationale

The four Docker services — Next.js, FastAPI, PostgreSQL + PostGIS, and background workers — together require approximately 2.5 GB RAM and 2 vCPU at idle. The 8 GB / 4-core VM provides comfortable headroom for portal traffic bursts and the monthly GEE batch ingest without requiring autoscaling infrastructure at pilot scale. Storage is sized at 30 GB for the VM disk (OS + Docker + live database); photos are stored in cloud object storage separately and are not counted against the VM.

Final resource limits are set at deployment time and adjusted from observed usage. A sizing review takes place at the end of Year 1 before any scale-up or handover decision.

8.2.3 Storage

The dominant storage item is photos from KoboCollect and WhatsApp submissions. Satellite-derived index values (Sentinel, CHIRPS) are written to PostgreSQL as aggregated rows — they produce negligible file storage. Where map overlay rasters are stored for the portal, they are kept at display resolution only — the resolution needed to render responsive portal tiles, not full satellite resolution.

Year 1 projection: well under 2 GB in cloud object storage. KoboCollect and WhatsApp compress photos at source (~100 - 200 KB each); at 32 sampling submissions and ~150 WhatsApp reports per month with photos, total photo storage is roughly 1GB by end of Year 1. Map overlay rasters at display resolution add a few MB. The VM's 30 GB disk covers the operating system, Docker images, and the live database only. Database backups go to cloud object storage separately — storing them on the same disk as the live data would defeat the purpose of a backup.

8.2.4 Resilience and Scaling

Resilience. Docker restarts failed containers automatically. If the VM itself fails, the platform is restored from the daily backup and redeployed from GitHub — target recovery within 4 hours (RTO). No data beyond the previous day's backup is lost (RPO 24 h).

Scaling after the pilot. The 8 GB / 4-core spec is the starting point, not the ceiling. If the Year 1 telemetry review shows sustained resource pressure, the next step is upsizing the VM — a cheap, reversible change with no code changes required.

8.3 Backup & Recovery

- Daily automated backups of all platform data
- Backup retention: 30 days

8.4 Monitoring & Alerting

- Infrastructure monitoring: server uptime, CPU and memory utilisation, disk usage
- Application monitoring: Africa's Talking webhook delivery failures, KoboToolbox sync errors, admin layer processing queue depth, portal availability
- Alerting: email notification to Akvo operations team on critical failures

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9. Testing & Validation Strategy

9.1 Technical Testing

- Unit testing: core business logic (health score calculation, validation rules, traffic light classification, data transformations)
- Integration testing: USSD session flows, WhatsApp Business bot flows, KoboToolbox API sync, Sentinel 1 & 2 and CHIRPS ingestion pipelines
- End-to-end testing: full wetland watcher journey (pollution episode) and full citizen scientist journey (monthly sampling)

9.2 Field Validation

- Field pilot with actual wetland watchers and citizen scientists before full rollout; document failure modes
- Train-the-Trainers (ToT) workshop as a system validation checkpoint — system must be usable by ToT-trained users from day one

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10. Project Timeline

Milestone	Date
Design signoff	2nd June 2026
Platform development	25 May – 26 June 2026
Training	End of June 2026
Data collection begins	Early July 2026

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11. Forms

Date source 1: Citizen reporting

- USSD based reporting

  Dial XXX to report a change in water colour, smell, fish or other animal kills, or change in water levels

Press 1 if the water colour suddenly became darker/murkier

Press 2: If the water is smelling bad

Press 3: If you see many dead fishes or animals in the river

Press 4: If the water is level is too high

Press 5: If the water level is too low after low flows

- Additional information in case of WhatsApp based reporting

  Take at least 2 photos of your observation

  Add voice notes/text comments if you deem necessary (Optional):

Data source 2: Citizen scientist (In Kobo)

Site details:

- Sampling Id
- Photo
- GPS reading
- Water quality
  - pH, Water Temperature and Dissolved Oxygen test results
    - Boundaries : pH= 2-10, Water Temp: 5-50 degree C, Dissolved Oxygen= 0.5- 35 mg/L
- Catchment
  - Type of crops grown in the catchment
  - Type of plant species in the catchment
- Ecological
  - % wetland site covered by invasive macrophytes
  - Catch per unit effort

        - Target species: Mainly, which fish species (X) are you capturing?
        - Total catch: How many (Y) of those species were caught?
        - Effort: How many hours did you spend actively fishing and getting the XY catch? (exclude travel/rest time)

- Hydrological
    - What is the water level during the time of sampling?
        - High (if the riverbanks are not visible and floodplains are inundated)
        - Medium (if the banks are visible and floodplains are not inundated)
        - Low (if the water has trickled down a small stream/narrow river and parts of the river bed are visible)

General comments

1. Consider all countries for citizen reporters
2. Consider other incidents or indicators of environmental degradation (check the TORs) for example soil erosion among others……
3. Put a list of acronyms for easy understanding for all.